METHOD FOR A PHYSICAL DATA ROUTING SERVICE

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Title: METHOD FOR A PHYSICAL DATA ROUTING SERVICE

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Abstract

A business method for providing a secure data transport service that leverages a physical logistics network where the data payload, not the original storage medium, is the routed entity. The method comprises: receiving a customer's dataset at a source hub by transferring it from a customer's portable storage medium to the internal storage of a first transport vehicle; routing the first transport vehicle to an intermediate or destination hub; optionally transferring the data payload to a second transport vehicle at an intermediate hub; and at the destination hub, transferring the data payload to a local portable medium for final delivery. This method provides a service analogous to packet-switching, where the data is the "packet" that moves through the network's physical infrastructure.

Background of the Invention

The transfer of massive datasets is a growing challenge for many industries. While internet backbone speeds have increased, they have not kept pace with the exponential growth of data. The concept of physically moving data has been hampered by slow I/O speeds, making

the process inefficient. A business model is needed that can offer a viable, high-speed, and secure alternative by treating physical data transport with the same sophistication as a digital network.

Summary of the Invention

The present invention is a business method for operating a secure, high-speed physical data routing service. The method redefines physical data transport from a simple courier service for a physical object to a sophisticated routing service for a data payload.

The core process involves offering a service where a customer's data payload is ingested at a source hub and transferred into the massive onboard storage of a network transport vehicle. The logistics provider then takes responsibility for routing the *data itself* to the destination. This may involve the vehicle traveling the entire distance or, more efficiently, involve the vehicle traveling to an intermediate hub where the data payload is rapidly transferred to another vehicle that is already heading toward the destination, much like a package at a sorting facility. At the destination hub, the data is offloaded to a "last-mile" portable cartridge for delivery. This business method creates a true physical data network, offering a service that is fundamentally different from simply transporting a hard drive.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a flowchart illustrating the steps of the physical data routing service, from data ingest to final delivery.

Detailed Description of the Invention

Referring to **FIG. 1**, the business method is detailed as a sequence of steps. The process begins when the service provider receives a data transport order (101). The customer's data payload is ingested at a source hub by transferring it from the customer's device to the onboard storage of a network transport vehicle (102). The provider's logistics system calculates an optimal physical route for the data payload (103). The vehicle then transports the data. At an intermediate hub, a routing decision is made (104). If the current vehicle is continuing to the destination, it proceeds. If transferring to a different vehicle is more efficient, the data payload is transferred to a new vehicle (105). This process repeats until the data arrives at the destination hub. There, the data is transferred to a last-mile delivery cartridge (106), which is delivered to the recipient, completing the service (107).

Claims

What is claimed is:

1. A method for providing a data routing service, the method comprising the steps of:

    - at a first physical hub, receiving a data payload by transferring it from a client's storage medium to an internal storage system of a first transport vehicle;
    - physically transporting the data payload to a second physical hub using the first transport vehicle;
    - at the second physical hub, transferring the data payload from the internal storage system of the first transport vehicle to a second storage system; and
    - billing a client for the routing of the data payload.

2. The method of claim 1, wherein the second storage system is an internal storage system of a second transport vehicle destined for a third physical hub.

3. The method of claim 1, wherein the second storage system is a portable storage medium for delivery to a final recipient.

4. The method of claim 1, wherein the step of transferring data is performed via a high-bandwidth parallel interface capable of a data transfer rate exceeding 1 terabit per second.

5. The method of claim 4, wherein the high-bandwidth parallel interface is a massively parallel optical interface.

6. The method of claim 1, wherein the data payload is encrypted, and wherein the service provider does not take possession of the decryption key.

Drawings

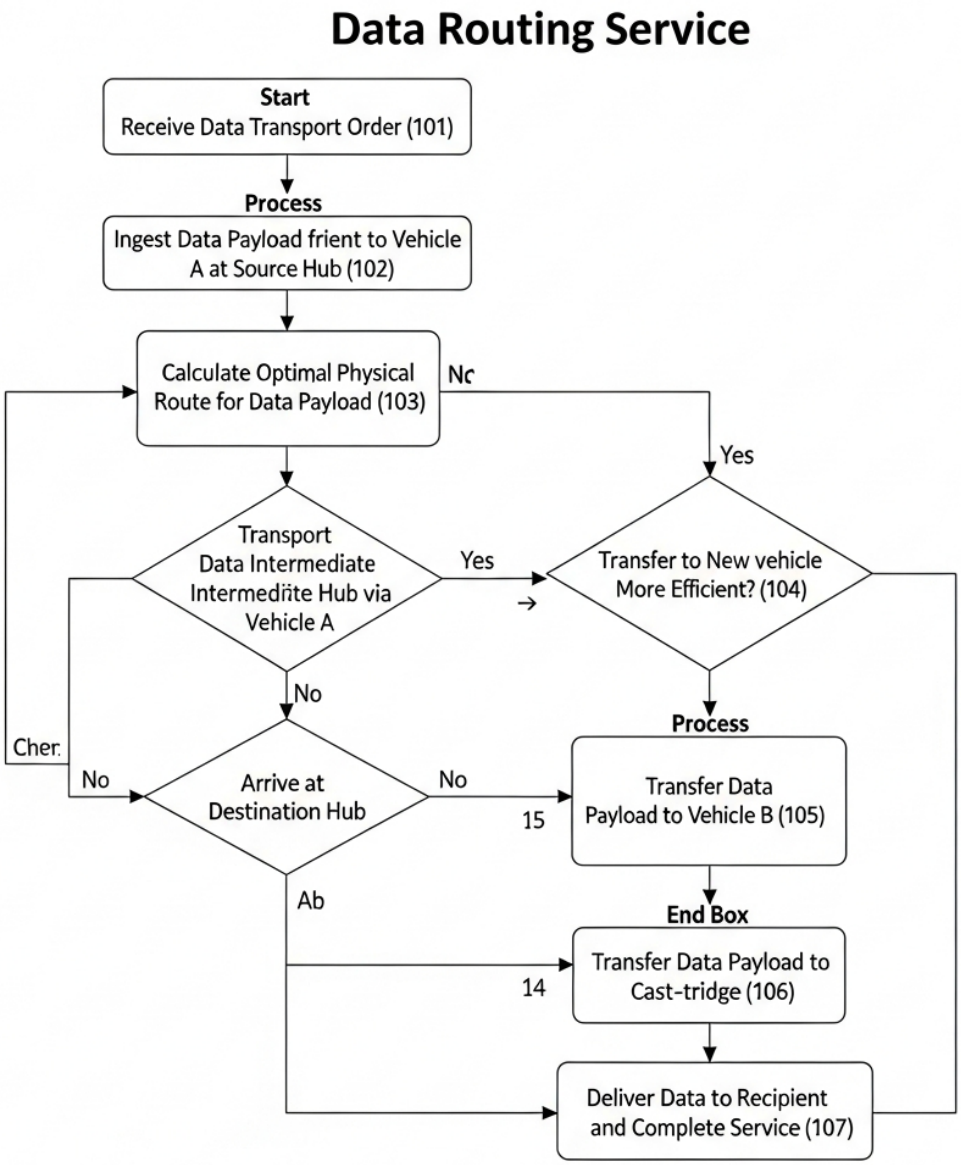


Figure 1: A flowchart illustrating the steps of the physical data routing service, from data ingest to final delivery.